

# Lean Six Sigma Black Belt Training

**Duration:** 1 Day

**Mode:** Online / Classroom / On-Site

**Certification:** QLSS Certificate of Completion



## Course Introduction

The Lean Six Sigma Black Belt Training is an advanced-level program that builds expertise in leading large-scale process improvement initiatives using complex analytical tools. Participants will learn to guide Green Belts, manage cross-functional projects, and deliver measurable business impact using Lean Six Sigma principles.

This program equips professionals with a deep understanding of statistical analysis, change management, project leadership, and strategic alignment of improvement efforts.



## Pre-requisites

- Prior Green Belt certification or equivalent project experience
- Working knowledge of DMAIC tools and basic statistics
- Suitable for experienced professionals, managers, and quality leaders



## Learning Objectives

- Lead complex Lean Six Sigma projects using advanced tools
- Mentor and coach Green Belts
- Apply advanced statistical methods for data-driven decisions
- Manage stakeholder expectations and change resistance
- Align improvement with organizational goals and ROI
- Build sustainable improvement systems



## Course Agenda

1. Role of the Black Belt
  - Strategic improvement alignment
  - Project selection and prioritization

## 2. Advanced Define & Measure

- QFD, Kano Analysis, and benchmarking
- Advanced MSA and reliability analysis

## 3. Analyze Phase – Advanced Tools

- Multiple regression, DOE (Design of Experiments)
- Non-parametric tests, correlation analysis

## 4. Improve Phase – Optimization

- DOE application, response surface methodology (RSM)
- Lean tools integration, mistake-proofing

## 5. Control Phase – Long-Term Control

- Advanced control charts (X-bar R, EWMA, CUSUM)
- Process audit, documentation, and institutionalization

## 6. Leadership and Change Management

- Building team dynamics and coaching Green Belts
- Leading cross-functional improvement workshops
- Overcoming resistance to change



### **What You'll Receive**

- QLSS Black Belt Certificate of Completion
- Training Material / Handout (PDF Format)
- Final Assessment and Evaluation Report



### **Who Should Attend**

- Lean Six Sigma Green Belts
- Quality and Process Excellence Professionals
- Operations Leaders and Department Heads
- Internal Trainers and Improvement Champions



## **Trainer Profile**

Our Master Black Belt trainers are seasoned professionals with a track record of delivering Lean Six Sigma transformations across Fortune 500 companies, MSMEs, and public sector organizations. They bring extensive project experience and coach Black Belts in practical application of tools and leadership behaviors for breakthrough performance.



## **Instructional Methodology**

- Simulation-based DMAIC workshops
- Advanced tool demonstrations using software (Minitab, Excel)
- Group coaching and case discussion
- Personalized project guidance (optional)



## **What Makes QLSS Black Belt Training Unique**

- Advanced-level statistics and project leadership focus
- Hands-on DOE, regression, and optimization exercises
- Strategic alignment with business goals and ROI delivery
- Mentoring support for post-training project execution